

- (c) Nepal, Pakistan, Bangladesh, China
(d) Pakistan, Nepal, China, Bangladesh

M.P. P.C.S. (Pre) 2019

Ans. (b)

The length of the international border (land border) with the neighbouring countries of India is as follows :

Country	Length of border (in km)
Bangladesh	4096.7
China	3488
Pakistan	3323
Nepal	1751
Myanmar	1643
Bhutan	699
Afghanistan	106
Total	15,106.7

Therefore, according to the question, the ascending order is - Nepal, Pakistan, China, and Bangladesh.

28. India's territorial limit extends towards the sea up to how many nautical miles/km from the coast?

- (a) 10 nautical miles (about 19.9 km)
(b) 16 nautical miles (about 25.9 km)
(c) 12 nautical miles (about 21.9 km)
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

India's territorial limit extends towards the up to 12 nautical miles (about 22.2 km) from the coast. Significantly, the mainland of India, extends from Kashmir in the north to Kanniyakumari in the south and Arunachal Pradesh in the east to Gujarat in the west. India's southern boundary extends up to 6° 45'N latitude in the Bay of Bengal. The actual distance measured from north to south is 3214 km and that of east to west is only 2933 km.

29. Which district of Uttarakhand is not situated along the Tibet boundary?

- (a) Uttarkashi (b) Chamoli
(c) Almora (d) None of the above

69th B.P.S.C. (Pre), 2023

Ans. (c)

Almora district of Uttarakhand is not situated along the Tibet boundary. Significantly, three districts of Uttarakhand share a border with Tibet-Chamoli, Pithoragarh and Uttarkashi.

2. Physical Divisions

i. Natural Regions of India

*Paleomagnetism is the branch that studies the magnetic properties preserved in the rocks, sediments or other such things during their formation. This branch of science is helpful in the study of ancient geological phenomena. **According to continental drift theory, the Indian mainland is a part of Gondwanaland.** Gondwanaland includes present day Peninsular India, Australia, Africa, Arabian Peninsula, South America and regions of Antarctica. *In the mid Mesozoic era (Jurassic Period, 144-208 Million Years Ago), Gondwanaland split into a number of pieces through convection currents. The Indian plate started drifting towards north after splitting from Gondwanaland. Consequently, the Indian plate, moving towards north, collided with a bigger plate, called the Eurasian plate.

Due to this collision, the sedimentary rocks of the Tethys Sea, between the two plates (i.e., Indian and Eurasian), folded and developed into the Himalayas and mountain ranges of West Asia. The whole mountain range of the Himalayas represents a youthful topography. It has high peaks, deep valleys (gorges), and fast-flowing rivers. In the Terai region of the Himalayas, **artesian wells** are also found.

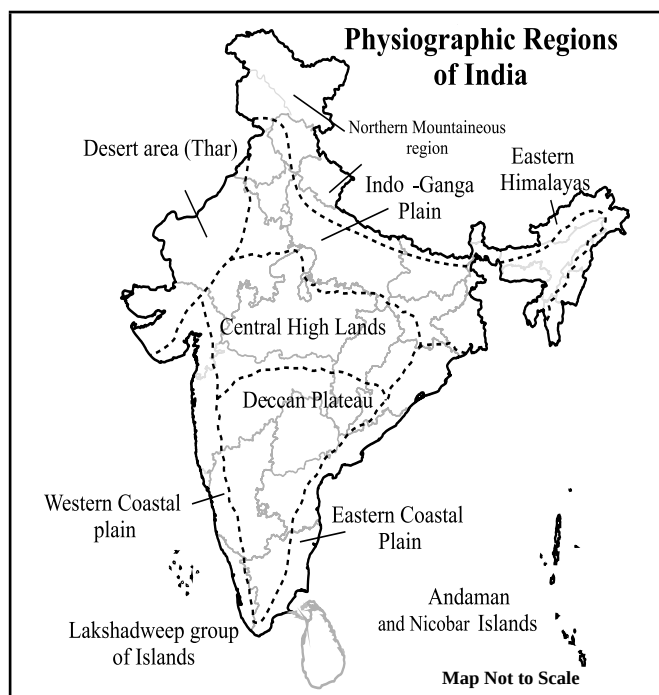
*Due to upliftment of **Tethys Sea** in the Himalayas and the subsidence of the northern flank of the peninsular plateau, resulted in the formation of a large basin. With the passage of time, this basin was gradually filled with sedimentary deposits of rivers flowing from Northern mountains and Peninsular Plateau. In this way, a vast land area of alluvial deposits was developed as the 'Northern Plain' of India.

*Geologically, Peninsular Plateau is the ancient part of Earth's surface. *Peninsular Plateau is the most stable region of India. Peninsular Plateau comprises of hills and broad valleys which are made up of igneous and metamorphic rocks. Meghalaya plateau is an extension of Peninsular plateau. It was separated from Indian Peninsula through **Malda Gap** due to faulting. The Western Coast of India was formed due to faulting and subsidence of the western flank of the plateau.

*The Indian mainland shows a large variation in the physical structure. Therefore India is divided into four natural regions. These are - 1. Northern Mountains 2. Northern Plains 3. Peninsular Plateau and 4. Coastal Plains & Islands. *Based on the variations in its geological structure and formations,

India can be divided into 3 primary and 6 sub-physical geomorphic regions-

1. The Northern and North-Eastern Mountains.
2. The Northern Plain
3. The Indian Desert
4. The Peninsular Plateau
5. The Coastal Plain
6. The Islands



1. The paleomagnetic results obtained from India indicate that in the past, the Indian land mass has moved :

- (a) Northward
- (b) Southward
- (c) Eastward
- (d) Westward

U.P.P.C.S. (Pre) 1998

I.A.S. (Pre) 1995

Ans. (a)

The Indian landmass has moved northward. This is evident from the paleomagnetic results obtained from India. Even the extension of the plateau region on the north side is evident. Paleomagnetism is the study of the record of the Earth's magnetic field in rocks, sediment or archaeological materials. This record provides information on the past behavior of Earth's magnetic field and the past location of tectonic plates. According to continental drift theory, Indian landmass was part of Gondwanaland, which included South America, Antarctica, Australia, Africa, The Arabian Peninsula and Peninsular India. This large landmass and broke Indian landmass moved northward. Vindhya and Western Ghats mountains emerged during this period.

2. The Indian subcontinent was originally part of a huge mass called :

- (a) Jurassic land mass
- (b) Aryavarta
- (c) Indiana
- (d) Gondwana Continent

I.A.S. (Pre) 1995

Ans. (d)

The Indian subcontinent was originally part of a huge landmass called Gondwana Continent. Gondwana was an ancient supercontinent that broke up about 200 million years ago. The continent eventually split into landmasses that we recognize today as Africa, South America, Australia, Antarctica, the Peninsular India and the Arabian Peninsula.

3. India is divided into how many Natural Territories ?

- (a) 4
- (b) 6
- (c) 7
- (d) 8

R.A.S./R.T.S. (Pre) 1996

Ans. (a)

India is divided into four main natural regions. These are –

- (i) Northern mountainous region
- (ii) Northern Plains
- (iii) Peninsular Plateau
- (iv) Coasts and Islands

It is further categorized into six sub-physical regions:

- (i) The Northern and North-Eastern Mountains
- (ii) The Northern Plain
- (iii) The Peninsular Plateau
- (iv) The Indian Desert
- (v) The Coastal Plains
- (vi) The Islands

4. Which one of the following pairs (Deccan Trap — Peculiarity) is not correctly matched?

- (a) Depth of middle trap — 1200 metres approximately
- (b) Intertrappean beds — Fossils of sea plants and animals
- (c) Depth of lower trap — 150 metres approximately
- (d) Depth of upper trap — 450 metres approximately

UPPSC (Pre) 2024

Ans. (b)

The Deccan Traps were formed in Western Central India during the Cretaceous period of the Mesozoic era, covering a large part of Maharashtra. Volcanic fissure eruptions led to lava flows that covered approximately 5 lakh square kilometers. In this region, basaltic lava deposits are found with thicknesses exceeding 2000 meters (reaching up to 3000 meters in some areas). The thickness of the Deccan Traps decreases towards the north, east and south. Intermediate layers of the traps do not contain fossils of marine plants and animals. Therefore, the statement given in option (b) is not correctly matched.

5. In which part of Uttarakhand are artesian wells found?

- (a) Bhabhar (b) Tarai
(c) Shivalik Hills (d) None of the above

Uttarakhand P.C.S. (Mains) 2006

Ans. (b)

An artesian well is simply a well that doesn't require a pump to bring water to the surface. This occurs when there is enough pressure in the aquifer. The pressure forces the water to the surface without any sort of assistance. In Uttarakhand, artesian wells are mostly found in the Tarai Region.

6. Consider the following statements regarding formation of landforms in India :

- I. Structurally, the Meghalaya plateau is an extended part of the Deccan plateau.
II. The Valley of Kashmir was formed in a synclinorium.
III. The Gangetic plain was formed in a foredeep.
IV. The Himalayas originated as a result of triangular convergence of the Indian plate, the European plate and the Chinese plate.

Which of these statements are correct?

- (a) I, II and III (b) I, III and IV
(c) I and III (d) II and IV

47th B.P.S.C. (Pre) 2005

Ans. (a)

Meghalaya plateau is an extension of the Peninsular plateau. It is believed that due to the force exerted by the northeastward movement of the Indian plate at the time of the Himalayan origin, a huge fault was created between the Rajmahal hills and the Meghalaya plateau. Later, this depression got filled up by the deposition activity of numerous rivers. Today, the Meghalaya and Karbi Anglong plateaus stand detached from the main Peninsular Block. Thus statement (1) is correct. The Valley of Kashmir was formed in a synclinorium. The Indo-Gangetic basin is an active foreland basin having an east-west

elongated shape. The basin formed in response to the uplift of the Himalaya after the collision of India and China plates Dewey and Bird (1970). Suess (1893-1909) was the first geologist to suggest that the Indo-Gangetic depression is a 'fore-deep' and was formed in front of the high crust-waves of the Himalayas as their southward migration was resisted by the rigid landmass of the Peninsula.

Thus, statement (3) is also correct. The Himalayan mountain range has formed as a result of the collision between the Indian plate and the Eurasian plate. The Chinese plate is not mentioned in the theory of plate tectonics. Thus statement (4) is wrong.

7. Which of the following statement is incorrect ?

- (a) Geologically Peninsula region is India's most ancient part.
(b) Himalaya mountains demonstrate the most newly developed folded mountains in the world.
(c) The Western coastline of India is formed by the deposition process of the rivers.
(d) Gondwana rocks have the largest reserve of coal in India.

U.P. Lower Sub. (Pre) 2004

Ans. (c)

The Western coastline of India is not formed by the deposition process of the rivers. In fact, it is formed by the submergence and emergence of the land. Gondwana rocks have the largest reserve of coal in India. The Himalaya is the newly developed folded mountain. Geologically, the Peninsular region is the most ancient part of India. Thus statement (c) is incorrect.

8. Match List I with List II and select the correct answer:

List-I	List-II
A. Deccan Traps	1. Late Cenozoic
B. Western Ghats	2. Pre-Cambrian
C. Aravalli	3. Cretaceous Eocene
D. Narmada-Tapti alluvial deposits	4. Cambrian
	5. Pleistocene

Code :

	A	B	C	D
(a)	3	5	1	4
(b)	3	1	2	5
(c)	2	1	3	4
(d)	1	4	2	5

I.A.S. (Pre) 1997

Ans. (b)

The correctly matched order is :

The origin of the Deccan Traps - Cretaceous Eocene
Origin of the Western Ghats - Late Cenozoic Era
Origin of the Aravalli Mountain - Pre-Cambrian Era
Narmada-Tapi alluvial deposits - Pleistocene Epoch

9. Kuttanad (or Kuttanadu) of Kerala is famous for :

- (a) A freshwater lake
- (b) The region with the lowest altitude in India
- (c) A coral island
- (d) The westernmost point of India

U.P.P.C.S. (Mains) 2015

Ans. (b)

Kuttanad (or Kuttanadu) of Kerala is famous for its lowest altitude in India. Kuttanad is called the Rice Bowl of Kerala. Here farming is practiced (4 to 10 ft.) 1.2 to 3.0 meters below sea level. FAO has declared the Kuttanad farming system as a globally important Agricultural Heritage System.

10. Which of the following is a desert region?

- (a) Sindhu area
- (b) Gangetic area
- (c) Assam area
- (d) Central India area

M.P. P.C.S. (Pre) 2018

Ans. (a)

The Sindhu area or the Indus Valley Region is a desert region located in the Northwestern Punjab Province (Pakistan).

11. Which one of the following districts does not have Dharwar geological formations?

- (a) Munger
- (b) Rohtas
- (c) Jamui
- (d) Nawada
- (e) None of the above/More than one of the above.

65th B.P.S.C. (Pre) 2019

Ans. (b)

Rohtas District does not have Dharwar geological formations. Districts having Dharwar land level composition in Bihar come under South-East Bihar. These are Munger, Nawada and Jamui districts. Whereas Rohtas is a part of South-West Bihar. Vindhyan type of ground level formation is found in this area. In Bihar, Vindhyan ground level formation is found in the Kaimur and Son river valleys (Rohtas).

12. Which of the following rock systems in India is the latest one?

- (a) Vindhyan
- (b) Cuddapah
- (c) Dharwar
- (d) Gondwana

U.P.P.C.S. (Pre) 2016

Ans. (d)

The rocks of the Dharwar order were formed by the transformation or faulting of the rocks of the Archean order. These rocks are highly metallic, in which minerals like gold, Iron, Manganese, Mica, Cobalt, Chromium, Copper, Tungsten, Lead, etc. are obtained. The Cuddapah Rock System was formed after the rocks of the Dharwar sequence. Limestone, Sandstone, Sugar clay, and Black soil are found in them. The rocks of the Gondwana sequence were formed long after the rocks of the Vindhya sequence. Most of India's coal is obtained from the Gondwana region only. The coal of this area is of high grade. Maximum Bituminous coal is obtained here.

13. Read the following statements and choose the correct option :

Statement - I : J.D. Dana prepared a special Geological Time Scale to indicate the geological history of Indian subcontinent rocks

Statement - II : As per Geological Time Scale, Indian rocks are subdivided into Archaean, Puranic, Dravidian and Aryan periods

- (a) Statement - I and Statement - II both are true
- (b) Statement - I and Statement - II both are false
- (c) Statement - I is true but Statement - II is false
- (d) Statement - I is false but Statement - II is true

Chhattisgarh P.C.S. (Pre) 2021

Ans. (d)

J.D. Dana's 'Manual of Geology' is a source of information regarding the geology of America. The term Archaean was introduced by J.D. Dana to designate the geological formations older than the Cambrian period. Sir T.S. Holland of the Geological Survey of India has classified the rock systems of the country into Archaean, Purana, Dravidian and Aryan periods.

14. Ganga Plain has been described as a :

- (a) pediplain
- (b) peneplain
- (c) geosyncline
- (d) karst plain
- (e) None of the above / More than one of the above.

67th B.P.S.C. (Pre) 2022

Ans. (c)